

MODULE DESCRIPTION FORM

نموذج وصف المادة الدراسية

Module Information			
معلومات المادة الدراسية			
Module Title	Programming Fundamentals		Module Delivery
Module Type	Core		☒ Theory ☒ Lecture ☒ Lab ☒ Tutorial ☒ Practical ☐ Seminar
Module Code	COMP101		
ECTS Credits	7		
SWL (hr/sem)	175		
Module Level	1	Semester of Delivery	
Administering Department	CS	College	CS and Mathematics
Module Leader	Names	e-mail	
Module Leader's Acad. Title	Assistant Professor	Module Leader's Qualification	Ph.D.
Module Tutor		e-mail	
Peer Reviewer Name	Name	e-mail	E-mail
Scientific Committee Approval Date	01/06/2023	Version Number	1.0

Relation with other Modules			
العلاقة مع المواد الدراسية الأخرى			
Prerequisite module	None	Semester	
Co-requisites module	Advanced Programming Language	Semester	2

Module Aims, Learning Outcomes and Indicative Contents

أهداف المادة الدراسية ونتائج التعلم والمحتويات الإرشادية

Module Objectives أهداف المادة الدراسية	1. This module aims to introduce the principles of Top Down problem solving strategy and algorithms 2. introduction to Java 3. Different data types are introduced to the student. 4. Java programming language (in NetBeans IDE environment) is adopted as a vehicle language for implementations. 5. Controlling the program flow using the loops and conditional statements
Module Learning Outcomes مخرجات التعلم للمادة الدراسية	1. Knowledge and understanding A student completing this module should be able to: 2. Understand the fundamental programming constructs. - Develop algorithms. - Understand, write and run basic java programs - Understand a typical Java program environment. 3. Cognitive skills (thinking and analysis). - To understand and analyze any problem and derive its solution. - To write a Java program for solving the problem. 4. Communication skills . - Write and present basic Java programs 5. Practical skills. Use the NetBeans IDE software to write, debug and run basic Java programs. 6. Job Skills - Be able to analyze the problems and convert the solutions into a programming code. - getting the required programming skills to be enrolled in the labor market
Indicative Contents المحتويات الإرشادية	The course focuses on problem solving strategies and the use of algorithmic language to describe such problem solving. It introduces the principles of procedural programming, data types, algorithms and problem solving methodologies. Introduction and history of Java, input output statements, identifiers, variables, arithmetic operators, math functions loops and conditional statements. All the programs will be written using Java programming language on NetBeans IDE 8.2 or higher.

Learning and Teaching Strategies

استراتيجيات التعلم والتعليم

Strategies	The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials, practical, lab hours and by considering types of some small projects that are interesting to the students.
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Student Workload (SWL)

الحمل الدراسي للطلاب محسوب لـ ١٥ أسبوعا

Structured SWL (h/sem) الحمل الدراسي المنتظم للطلاب خلال الفصل	110	Structured SWL (h/w) الحمل الدراسي المنتظم للطلاب أسبوعيا	7
Unstructured SWL (h/sem) الحمل الدراسي غير المنتظم للطلاب خلال الفصل	65	Unstructured SWL (h/w) الحمل الدراسي غير المنتظم للطلاب أسبوعيا	6
Total SWL (h/sem) الحمل الدراسي الكلي للطلاب خلال الفصل	175		

Module Evaluation

تقييم المادة الدراسية

		Time/Number	Weight (Marks)	Week Due	Relevant Learning Outcome
Formative assessment	Quizzes	2	10% (10)	5 and 10	All
	Assignments	2	10% (10)	4 and 12	All
	Projects / Lab.	1	10% (10)	Continuous	All
	Report	1	10% (10)	13	All
Summative assessment	Midterm Exam	2hr	10% (10)	7	All
	Final Exam	3hr	50% (50)	16	All
Total assessment			100% (100 Marks)		

Delivery Plan (Weekly Syllabus)

المنهاج الاسبوعي النظري

	Material Covered
Week 1+2	Introduction: What is programming? Programming Languages, Types Programming Languages, Algorithms, Java, the World Wide Web, and Beyond ...
Week 3+4	Java program structure: Identifier, Variable, Data Types in Java, arithmetic operators and Java Type Casting
Week 5+6	Decision-Making statements: If Statement, if-else statement, if-else-if ladder and Nested if-statement
Week 7	First Exam
Week 8	Java Switch Statement
Week 9	Loops in Java: Java Simple Loop (For loops)
Week 10	Loops in Java: nested Loops (For loops)
Week 11	Loops in Java: While and do-while loops
Week 12	Applications on Loops
Week 13	Second Exam
Week 14	Revision and project discussion
Week 15	Final Exam

Delivery Plan (Weekly Lab. Syllabus)

المنهاج الاسبوعي للمختبر

	Material Covered
Week 1	Lab 1: Introduction to Java environment (NetBeans IDE)
Week 2	Lab 2: Basic Data types
Week 3	Lab 3: Program Structure and write small programs and running
Week 4	Lab 4: Write programs with arithmetic operators
Week 5	Lab 5: if statements, if else statements
Week 6	Lab 6: if-else-if ladder and nested if statements
Week 7	Lab 7: Applications on if statements
Week 8	First Exam
Week 9	Lab 8: Loops
Week 10	Lab 9: nested Loops
Week 11	Lab 10: While and do-while loops (application of the difference)
Week 12	Lab 10: Second Exam
Week 13	Lab 11: Write java programs in assignments
Week 14	Lab 12: Discussion assignments
Week15	Final Exam

Learning and Teaching Resources		
مصادر التعلم والتدريس		
	Text	Available in the Library?
Required Texts	Java How to Program (9 th Edition) ,Paul and Harvey Deitel,2011.	Yes
Recommended Texts	Java for Absolute Beginners, Iuliana Cosmina, 2018	Yes
Websites	☐ https://www.w3schools.com/java/ ☐ https://www.codecademy.com/learn/learn-java ☐ https://www.javatpoint.com/java-tutorial	

Grading Scheme				
مخطط الدرجات				
Group	Grade	التقدير	Marks %	Definition
Success Group (50 - 100)	A - Excellent	امتياز	90 - 100	Outstanding Performance
	B - Very Good	جيد جدا	80 - 89	Above average with some errors
	C - Good	جيد	70 - 79	Sound work with notable errors
	D - Satisfactory	متوسط	60 - 69	Fair but with major shortcomings
	E - Sufficient	مقبول	50 - 59	Work meets minimum criteria
Fail Group (0 – 49)	FX – Fail	راسب (قيد المعالجة)	(45-49)	More work required but credit awarded
	F – Fail	راسب	(0-44)	Considerable amount of work required
Note: Marks Decimal places above or below 0.5 will be rounded to the higher or lower full mark (for example a mark of 54.5 will be rounded to 55, whereas a mark of 54.4 will be rounded to 54. The University has a policy NOT to condone "near-pass fails" so the only adjustment to marks awarded by the original marker(s) will be the automatic rounding outlined above.				